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Genome-Wide analysis of the *NRAMP* gene family in *Arabidopsis thaliana*: identification, expression and response to multiple heavy metal stresses and phytohormones

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Abstract
Heavy metal stress is a critical challenge to agricultural productivity, necessitating deeper insights into the molecular mechanisms of metal transport in plants. In this study, we conducted a comprehensive genome-wide characterization of the Natural Resistance-Associated Macrophage Protein (NRAMP) gene family in *Arabidopsis thaliana* and identified six *AtNRAMP* genes. Phylogenetic and synteny analyses revealed their distribution into two distinct clades and evolutionary conservation with legumes such as *Glycine max* and *Arachis hypogaea*, indicating functional divergence and gene duplication events maintained under purifying selection. Conserved protein motifs and domains, particularly the NRAMP transmembrane domain, highlighted their conserved role in divalent metal ion transport, while cis-regulatory element analysis demonstrated enrichment of stress- and hormone-responsive elements, pointing to tight transcriptional regulation under environmental challenges. Structural modeling further supported the functional conservation of AtNRAMP proteins. Expression profiling showed clear tissue-specific expression under normal conditions and strong, differential regulation in response to cadmium and other heavy metals, as well as to the phytohormone abscisic acid (ABA). Collectively, these results provide foundational insights into the evolutionary relationships, regulatory mechanisms, and stress-responsive expression of the AtNRAMP gene family, offering a framework for future functional studies and potential applications in developing crops with enhanced heavy metal tolerance and improved growth under stress conditions.

Keywords Arabidopsis thaliana, *NRAMP* genes, Heavy metal stress, Gene duplication, Cis-regulatory elements, Phytohormones, Growth and development

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Introduction

Feeding an exponentially growing global population in the era of global climate change is a major challenge for sustainable agriculture. Among various environmental constraints, abiotic stressors are among the leading causes of reduced crop productivity and food insecurity [1, 2]. Heavy metal (HM) toxicity is particularly concerning due to its adverse effects on soil health, plant physiology, and ultimately human nutrition. HM accumulation disrupts soil pH, alters nutrient balance, and contributes to long-term soil degradation [3, 4]. While some HMs, such as iron, zinc, and manganese, are essential cofactors in plant metabolism, non-essential metals like cadmium (Cd) are toxic even at very low concentrations [5]. Cd is considered the most hazardous among heavy metals due to its high solubility, mobility, and persistence, as well as its tendency to enter plants through nutrient transport pathways, particularly those used by iron [6–8]. Its accumulation in edible plant tissues not only compromises crop quality but also poses severe health risks to humans, making Cd a critical global concern associated with industrialization, mining, and intensive agricultural practices [8–10].

Plants have developed a wide array of adaptive strategies to counteract Cd-induced stress, including antioxidant defense systems, vacuolar sequestration, and the regulation of metal transporters. In *Arabidopsis thaliana*, for instance, heavy metal transporting ATPases such as *AtHMA3*, members of the ABC transporter family like *AtMRP1*, *AtMRP2*, and *AtPDR8*, and transporters from the *NRAMP* family all play crucial roles in detoxification, sequestration, and efflux of Cd [11–13]. Beyond transporter activity, Cd exposure induces physiological disruptions, including inhibition of root meristem growth via nitric oxide (NO) signaling and perturbations in Ca^{2+} influx, which interfere with root hair elongation and overall plant development [14, 15]. Together, these findings highlight the complexity of Cd toxicity in plants and the central role of molecular transport systems in mediating tolerance.

In addition to transporters, phytohormones have emerged as environmentally friendly regulators of HM stress tolerance. These signaling biomolecules, produced in minute quantities, modulate diverse physiological and biochemical processes critical for plant survival [16, 17]. Phytohormones stimulate various signaling cascades that help mitigate HM-induced cellular damage, and both foliar applications and shifts in endogenous phytohormone levels under HM stress have been shown to improve plant tolerance and adaptation [18–22]. Recent advances in research tools such as genomics, transcriptomics, proteomics, and systems biology have enabled the characterization of key biomolecular pathways involved in phytohormone-mediated HM stress responses [23, 24]. These insights suggest that phytohormones act in close coordination with metal transport systems, providing plants with a multi-layered defense strategy against heavy metal stress.

Within this broader network, natural resistance-associated macrophage proteins (*NRAMPs*) stand out as one of the most evolutionarily conserved transporter families, mediating the transport of divalent metal ions across membranes. *NRAMPs*, first identified in mice, are now well established in plants, fungi, and other organisms as vital players in metal homeostasis and stress responses [25–27]. In plants, they facilitate the uptake and mobilization of Fe^{2+}, Mn^{2+}, and Cd^{2+}, though selectivity varies among isoforms. In *A. thaliana*, *AtNRAMP1* is known for high-affinity Mn uptake and Fe transport [28–30], while *AtNRAMP6* contributes to Fe homeostasis in Arabidopsis [31]. NRAMP homologs have also been identified in rice, soybean, tomato, kiwifruit, Aegilops tauschii, Sorghum bicolor, Wheat, and Brassica species, where they show diverse roles in nutrient acquisition and stress adaptation [32–39]. Comparative genomic studies have revealed substantial variation in *NRAMP* gene families across species, with expression analyses showing that many members are responsive to Cd and other heavy metals, such as *BnNRAMP5.1/5.2* in *Brassica napus* and *NRAMP* groups in tea plants [34, 40].

Despite these advances, a systematic genome-wide investigation of *NRAMP* genes in *Arabidopsis thaliana* remains limited. Such analysis is crucial to better understand their evolutionary relationships, structural features, regulatory elements, and roles under heavy metal and phytohormonal stress. To address this gap, the present study undertakes a comprehensive genome-wide characterization of *NRAMP* family members in *A. thaliana*. The specific objectives are to (i) identify and classify *NRAMP* genes, (ii) analyze their expression patterns under heavy metal and phytohormone treatments, and (iii) examine their conserved motifs, cis-regulatory elements, protein domains, chromosomal distribution, duplication events, and synteny analysis. Furthermore, physicochemical characterization and 3D structure prediction were performed to infer their potential functional roles. This integrative approach is expected to enhance our understanding of NRAMP-mediated metal transport, their interaction with phytohormonal signaling, and their contribution to plant adaptation under heavy metal stress.

Materials and methods

Acquisition of *AtNRAMP* gene sequences and conserved domain analysis

To execute a thorough genome-wide assessment of *NRAMP* genes in Arabidopsis, the sequences of 6 *AtNRAMP* genes were downloaded from the Phytozome v13 database. Following examining *NRAMP* genes in Arabidopsis, the genomic information encompassing coding sequences (CDS), the genomic sequences of *AtNRAMP* genes, and in the same way Protein sequences were also retrieved from Ensembl Plant database v60 (2024) (https://plants.ensembl.org/index.html). All selected sequences were confirmed to possess the NRAMP domain using the Simple Modular Architecture Research Tool (SMART) database (http://smart.embl-heidelberg.de/). Further gene structure containing positions of intronic and exonic regions was examined using the GSDS 2.0 database of all 6 *AtNRAMPs*. Following initial screening using PFAM, potential *NRAMP* genes in Arabidopsis were verified against the Conserved Domain Database (CDD) (http://smart.embl-heidelberg.de/) to guarantee the existence of the distinctive NRAMP domain (PF01566). Protein sequences that included the NRAMP domain were extracted and used for additional bioinformatic analysis.

Analysis of conserved motifs and cis-regulatory elements

To elucidate the promoter regions of AtNRAMPs and their regulatory mechanisms, sequences spanning 1000 bp upstream were procured from the Ensemble Plants database v60 (2024) and subjected to analysis via the PlantCARE Database (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/) [41] for the examination of cis-elements. A thorough investigation of the PlantCARE database revealed 45 unique stress-response cis-regulatory elements. Every AtNRAMP gene's promoter's frequency of each cis-element was measured. To provide a comparative scale, these frequency data were then log-transformed and normalized (minimum = 1, maximum = 5). TBtools-II v2.310 software was used to create a heatmap visualization of the results. The heatmap provides a comparative insight of the regulatory potential of the six AtNRAMP promoters by displaying the relative presence of cis-elements across them. The identification of protein motifs was conducted using the MEME Suite, the MEME Suite (https://meme-suite.org/meme/) was set to detect a maximum of ten conserved motifs, while all other parameters were maintained at their default configurations. Complete gene structure illustration of AtNRAMP genes was done by TBtools-II v2.310 software [27].

Evolutionary analysis of NRAMP proteins across crop species

To understand the phylogenetic relationship between *Arabidopsis thaliana* and other species with known *NRAMP* genes, their full-length protein sequences were retrieved from the Phytozome database. Phylogenetic relationships among NRAMP proteins from *Arabidopsis thaliana* (6 *AtNRAMPS*), *Triticum aestivum* (13 *TaNRAMPs*), *Solanum tubersom* (5 *StNRAMPs*), *Oryza sativa* (7 *OsNRAMPs*), *Glycine max* (10 *GmNRAMPs*), *Arachis hypogaea* (14 *AhNRAMPs*), *Zea mays* (7 *ZmNRAMPs*), and *Camellia sinensis* (11 *CsNRAMPs*) were examined by constructing a phylogenetic tree with the neighbor-joining method, using 1000 bootstrap replicates and amino acid sequences from these species. The ClustalW algorithm in MEGAX software was used for alignment, and the tree was visualized using iTOL software (https://itol.embl.de/).

Chromosomal localization, gene duplication, and synteny analysis

The chromosomal positions of AtNRAMP genes were determined using coordinates extracted from the Ensembl Plants database and visualized with PhenoGram (http://visualization.ritchielab.org/phenograms/plot). The Circos software displayed gene-chromosome associations through its mapping feature, and TBtools v1.120 [27] produced the chromosome position map. Pairwise distances were computed using MEGA X to determine synonymous (Ks) and non-synonymous (Ka) substitution rates to evaluate evolutionary pressure. While a synonymous replacement (Ks) does not modify the amino acid sequence, a non-synonymous substitution (Ka) is a nucleotide alteration that modifies the encoded amino acid. Ks values in protein-coding genes usually surpass Ka values because, in accordance with the neutral theory of molecular evolution, synonymous substitutions are frequently tolerated while purifying selection eliminates the majority of nonsynonymous changes [42]. Selection pressure was measured using the Ka/Ks ratio, where purifying selection is indicated by a Ka/Ks < 1 ratio, neutral evolution is suggested by a Ka/Ks = 1 ratio, and positive selection is indicated by a Ka/Ks > 1 ratio. Ka and Ks scores were then used to identify tandem and segmental duplicates. A synteny relationship analysis between *AtNRAMPs* alongside various species was performed with the software program MCscanX inside Tbtools. The following parameters were set for MCScanX: alignment method = BLASTP, maximum number of BLAST hits retained per gene = 5, E-value threshold = $1e^{-10}$, and number of CPUs = 2. The Tbtools-II program facilitated the creation of this multiple synteny analysis. The analysis of synonymous and non-synonymous substitutions was performed with MEGAX to determine pairwise distances and subsequent detection of tandem and segmental duplicates through Ka and Ks scores.

Physicochemical properties, 3D model prediction of AtNRAMPs and Protein-Protein interaction

The ProtParam tool on the ExPASY website (https://web.expasy.org/protparam/) and Ensembl Plants were used to analyze the chromosomal number, molecular weight (MW), isoelectric point (PI), total counts of positively and negatively charged residues, grand average of hydropathicity (GRAVY), and TMDs in AtNRAMP proteins. The subcellular localization of AtNRAMP proteins was predicted using the WoLF PSORT online platform (https://wolfpsort.hgc.jp/). Protein-protein interaction networks were established using the STRING v12.0 database (https://string-db.org/), with interactions analyzed based on protein nomenclature and sequences. 3D-homology modeling of AtNRAMP proteins was done using SwissModel.

Plant materials, growth and treatments conditions

Seeds of wild-type *Arabidopsis thaliana* Columbia (Col-0) were originally sourced from the Arabidopsis Biological Resource Center (ABRC) and were obtained during doctoral studies from Professor Men's laboratory at Nankai University, Tianjin, China. Seeds were surface sterilized with 70% ethanol for 5 min, followed by 1% Clorox bleach for 10 min, then stratified at 4 °C for three days. Sterilized seeds were sown on solid Murashige and Skoog (MS) medium and grown vertically at 22 °C under a 16-hour light/8-hour dark photoperiod. For abscisic acid (ABA) treatment assays, seeds were sterilized and stratified as described in [43], then plated on MS medium supplemented with or without 5 µM ABA.

For heavy metal stress experiments, stratified seedlings were initially grown on MS plates for seven days before being transferred to pots containing a 3:1 mixture of soil and vermiculite [43]. Plants were grown for an additional four weeks, with 09 plants per pot and a total of 108 plants per stress in each independent experiment. After four weeks, uniformly healthy plants were randomly divided into five groups. For cadmium (Cd) treatments, plants were exposed to increasing concentrations of Cd^{2+} (10 mM) by applying 15 mL of $CdCl_2$ solution to the soil every two days [44]. For subsequent copper (Cu), (C) lead (Pb), and (D) zinc (Zn) stresses, we followed the protocols described in [33]. Control plants received an equal volume of deionized water. After seven days of treatment, tissue samples were harvested, immediately frozen in liquid nitrogen, and stored at −80 °C for subsequent analyses. For expression analysis, leaf samples from *Arabidopsis* plants were collected and used for subsequent experiments. All experiments were conducted with three independent biological replicates.

RNA extraction and QRT-PCR analysis

qRT-PCR was conducted to assess the transcriptional levels of the *six AtNRAMP* genes under multiple heavy metals and hormones. Total RNA was isolated from 7-day-old seedlings using the Trizol reagent (TransGen Biotech, Beijing, China). First-strand cDNA synthesis was carried out using the EasyScript First-Strand cDNA Synthesis SuperMix (TransGen Biotech, Beijing, China). The *ACTIN2* gene served as an internal reference for normalization. Primer sequences used for amplification are provided in Supplementary Table S1.

Tissue Specific and Responses of Multiple Heavy Metals Stresses and Hormone Treatment Expression Profiling of *AtNRAMPs* Family

For qRT-qPCR analysis, of different heavy metals stress and hormones, we followed previously described protocol in [43].

Results

Characterization and duplication of *AtNRAMPs*

Six *AtNRAMPs* were identified and studied in this study. Their physicochemical properties were determined by ProtPram ExPASY CELLO v2.5 [45] and are given in Table 1. According to these parameters, all protein lengths ranged between 509 aa to 532 aa, and their molecular weight were between 56.13 kDa (*AtNRAMP3*) to 58.7 kDa (*AtNRAMP5*). All *AtNRAMPs* showed PI range from 4.99 to 8.86, and their GRAVY values were all found to be positive, showing their high hydrophobicity. All 6 *AtNRAMPs* were found to have 12 TMDs.

AtNRAMPs were further analyzed for duplication events. The sequences were aligned in MEGAX, and pairwise distance was computed to find synonymous (Ks) and non-synonymous (Ka) substitutions. From these substitutions, 9 duplicate pairs were identified, among which one was a Tandem duplicate and the remaining 8 were segmental duplicates (Table 2). Ka and Ks values were further used to find Ka/Ks, and all pairs showed (Ka/Ks < 1), which means all these duplication events

Table 1 Physiochemical properties of *AtNRAMPs*

Proteins	No of amino acids	Molecular Weight (kDa)	PI	Negative Residues	Positive Residues	GRAVY	TMDs	Subcellular localization
AtNRAMP1	532	57.56	8.86	31	39	0.663	12	Plasma membrane
AtNRAMP2	530	58.42	5.06	47	36	0.46	12	Plasma membrane
AtNRAMP3	509	56.14	5.11	44	32	0.55	12	Plasma membrane
AtNRAMP4	512	56.39	4.99	48	36	0.554	12	Plasma membrane
AtNRAMP5	530	58.78	4.84	50	33	0.419	12	Plasma membrane
AtNRAMP6	527	57.25	8.03	34	36	0.602	12	Plasma membrane

Table 2 Ka/Ks of *AtNRAMPs* and duplication events

NRAMPs	Gene Name	NRAMPs	Gene Name	Ks	Ka	Ka/Ks	Duplication Type
AtNRAMP1	AT1G80830.1	AtNRAMP3	AT2G23150.1	2.071	0.664	0.320619	Segmental
AtNRAMP1	AT1G80830.1	AtNRAMP6	AT1G15960.1	0.747	0.073	0.098334	Tandem
AtNRAMP2	AT1G47240.1	AtNRAMP3	AT2G23150.1	1.665	0.176	0.105729	Segmental
AtNRAMP2	AT1G47240.1	AtNRAMP4	AT5G67330.1	2.246	0.199	0.088431	Segmental
AtNRAMP2	AT1G47240.1	AtNRAMP5	AT4G18790.1	2.418	0.190	0.078545	Segmental
AtNRAMP3	AT2G23150.1	AtNRAMP4	AT5G67330.1	1.929	0.163	0.084745	Segmental
AtNRAMP3	AT2G23150.1	AtNRAMP5	AT4G18790.1	1.534	0.255	0.166397	Segmental
AtNRAMP3	AT2G23150.1	AtNRAMP6	AT1G15960.1	1.680	0.639	0.380389	Segmental
AtNRAMP4	AT5G67330.1	AtNRAMP5	AT4G18790.1	2.168	0.257	0.11875	Segmental

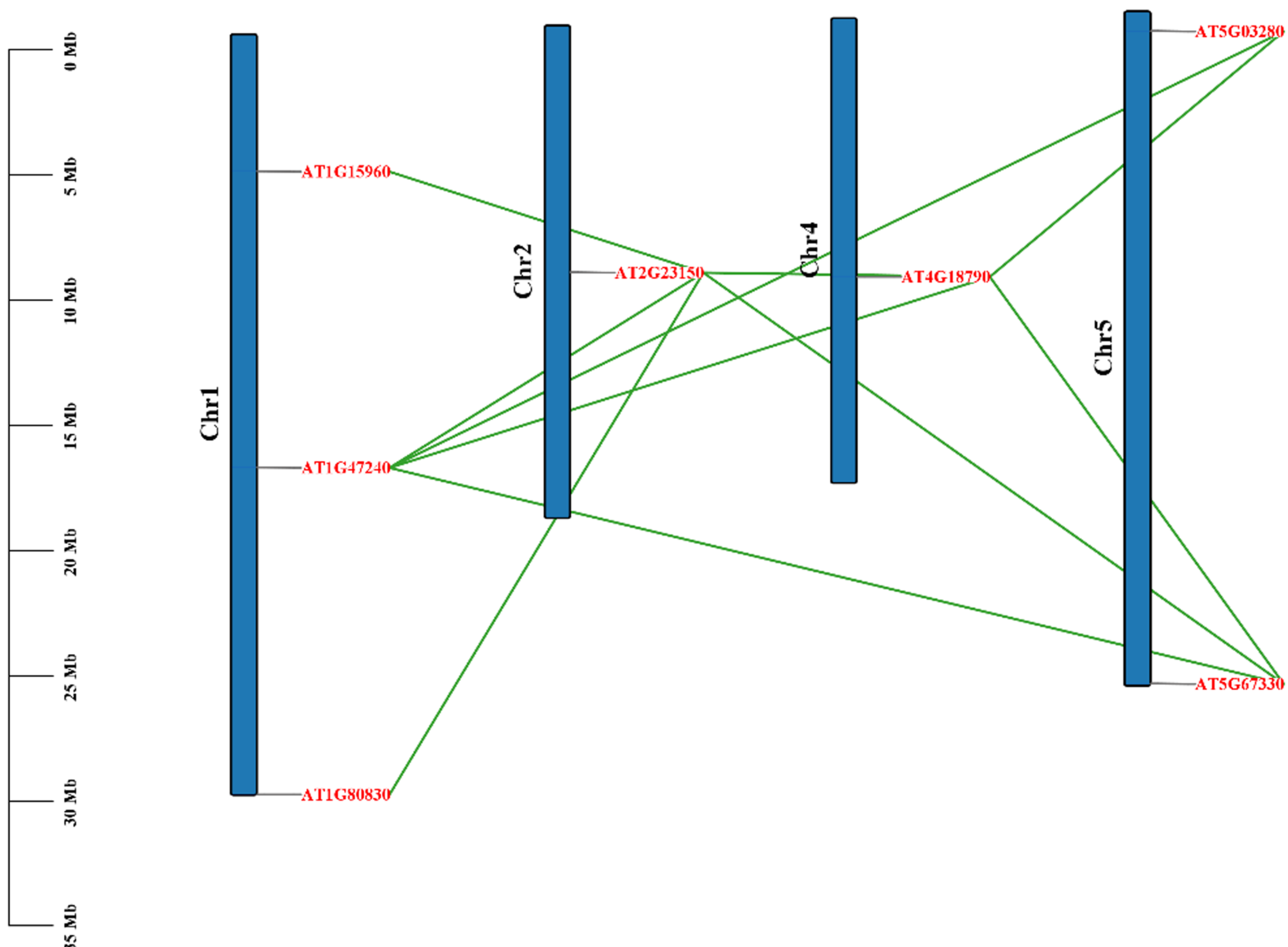

Fig. 1 *AtNRAMPs* gene location and duplicates. Chromosomes are shown in blue color. *AtNRAMPs* are shown with red color and green lines indicate the duplicated genes

happened under highly purifying natural selection. All of these duplicate pairs of chromosomes are shown in Fig. 1.

AtNRAMPs motif, domains, and gene structure analysis

AtNRAMPs phylogenetic analysis divided the six genes into two clades. Clade 1 has *AtNRAMP3*, *AtNRAMP2*, and *AtNRAMP4*, while clade 2 has *AtNRAMP1*, *AtNRAMP5*, and *AtNRAMP6*, respectively, showing a close evolutionary relationship among these two pairs of triplet *AtNRAMPs* (Fig. 2A). These *NRAMPs* were further analyzed in this study to understand the gene structure as paralogs tend to have less conservation in their genetic structure.

Gene structure analysis revealed that lowest number of introns (two) and exons (three) were present in *AtNRAMP4* of clade 1 and other two members of this clade (*AtNRAMP2* and *AtNRAMP3*) had 3 introns and 4 exons each. All three members of clade 1 had 2 UTRs each. In

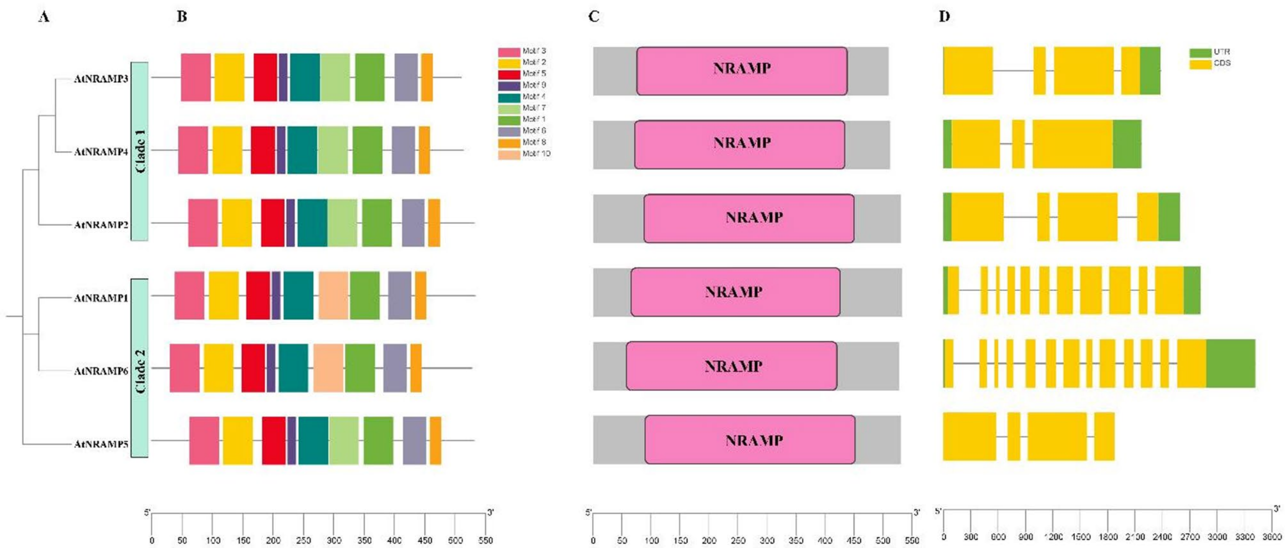

Fig. 2 *AtNRAMPs* characterization and conservation. (A) *AtNRAMPs* phylogenetic analysis by MEGAX showing two clades (B) Motiff analysis among 6 *AtNRAMPs* showing 10 conserved motifs (C) NCBI-CDD domain analysis showing *NRAMP* (Pink) domain in all *AtNRAMPs* (D) Exon/Intron distribution pattern with exons (yellow), introns (gray lines), and UTRs (green). The final image constructed in TbTools

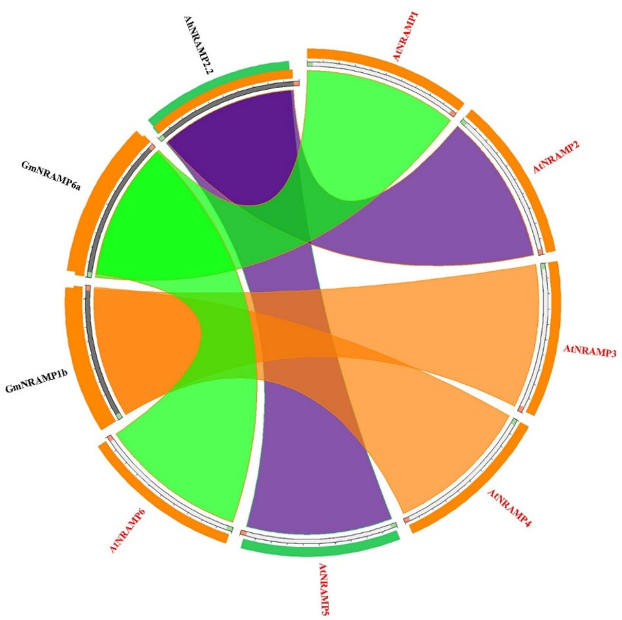

Fig. 3 Circoletto plot of NRAMP orthologs. Colored ribbons show percentage identity

clade 2, *AtNRAMP5* had 3 introns and 4 exons with no UTRs. *AtNRAMP1* and *AtNRAMP6* had maximum numbers of introns and exons showing different exon intron arrangement then rest of the members. *AtNRAMP1* had 11 exons and 10 introns with 2 UTRs. Lastly, *AtNRAMP1* had 13 exons and 12 introns with 2 UTRs (Fig. 2D).

All AtNRAMPs proteins were further analyzed for the presence of conserved motifs and domains to better understand the function of these proteins. All AtNRAMPs had NRAMP transmembrane domain which is crucial for transport of divalent metal ions. All *AtNRAMPs* had this domain between 50 and 450 bp regions (Fig. 3C). These proteins were further analyzed by MEME tool and about 10 conserved motifs were found in these *AtNRAMPs*. Among these, motif 1,3,2,5,6,8,9, and 4 were found in all *AtNRAMPs*. Motif 7 was found in four proteins except *AtNRAMP1* and *AtNRAMP6*. Instead of motif 7, *AtNRAMP1* and *AtNRAMP6* had motif 10 (Fig. 3B). Presence of almost similar motifs in all *AtNRAMPs* showed how much these genes are conserved functionally. Also, the presence of motif 10 in only two proteins is also important in unrevealing the functions of these two *AtNRAMPs*.

Phylogenetic and evolutionary relationships among *AtNRAMP* orthologs

To study the phylogenetic relationship between members of the *NRAMPs* gene family, a phylogenetic tree was constructed. The tree was Made using protein sequences of 6 plant species in which *NRAMP* genes have been identified. These include *Arabidopsis thaliana* (Thale cress), *Triticum aestivum* (Bread wheat), *Solanum tuberosum* (Potato), *Glycine max* (Soy bean), *Oryza sativa*, *Arachis hypogaea* (Peanut), *and Camellia sinensis* (Tea plant). All the protein sequences were used to build a multiple sequence alignment following a phylogenetic tree (Fig. 4).

The resulting tree was divided into 3 clades. Clade 1 contained no *AtNRAMP* genes, and it mostly has *TaNRAMPs* and *OsNRAMPS* in the majority. Three NRAMPs, each from *A. hypogaea*, *Z. mays*, and *C. sinensis*, were also in this clade, followed by 1 *StNRAMP*. Clade 2 contained 2 *AtNRAMPs* (*AtNRAMP6* and *AtNRAMP1*) which showed close phylogenetic relationship

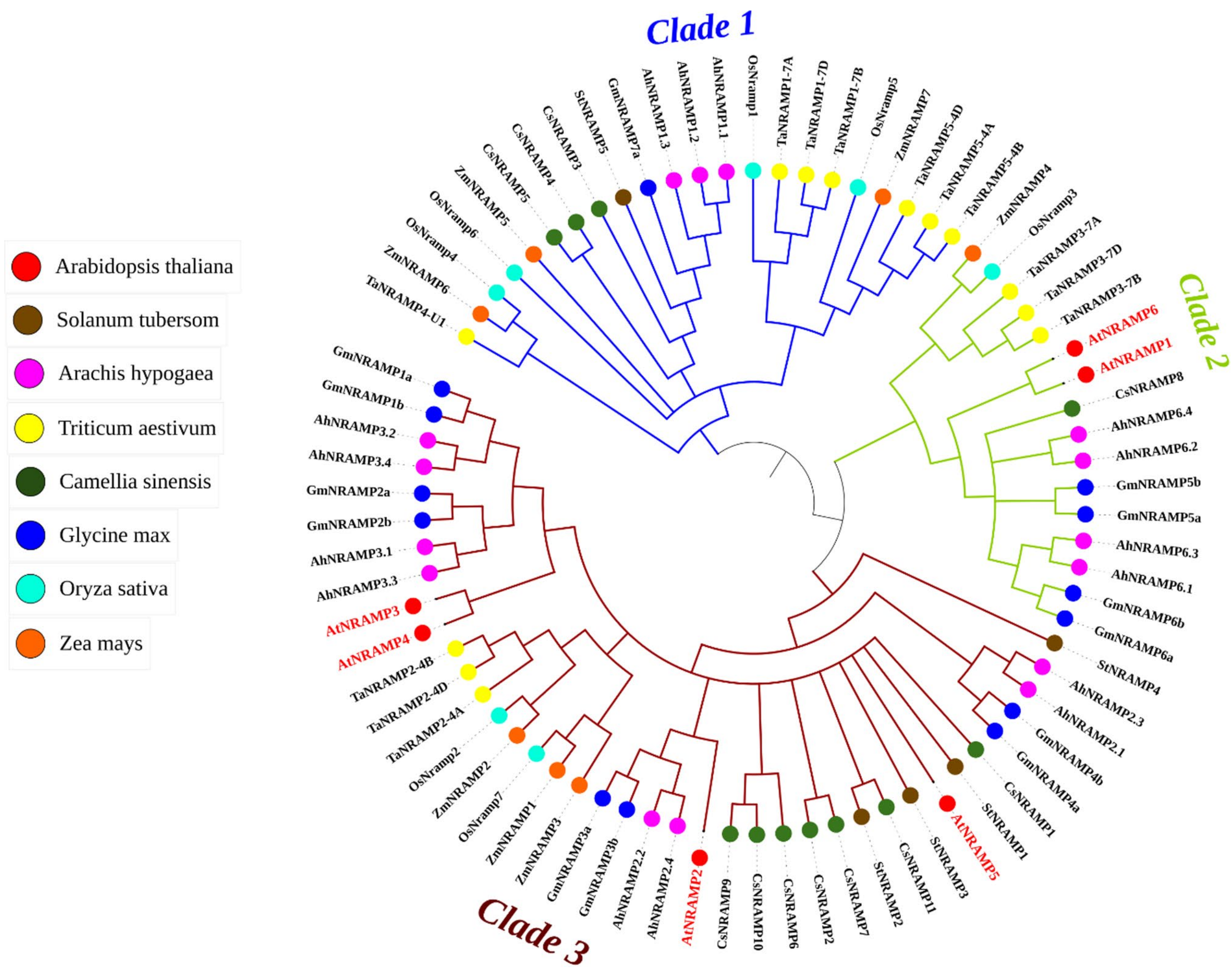

Fig. 4 Phylogenetic analysis among *NRAMP* genes from *A. thaliana* and 7 other species. The tree was annotated using iTOL and NJ method, with 1000 bootstrap replications used in MEGA for phylogenetic analysis. The tree shows the division of *NRAMPs* into 3 clades. *AtNRAMPs* are shown in red color, and different colored dots represent different species as mentioned in the legend

with *OsNRAMP3*, *AmNRAMP4*, *TaNRAMP3-7 A*, and *TaNRAMP3-7B*, *TaNRAMP3-7D*. Clade 3 was the largest and contained 4 AtNRAMPs (*AtNRAMP2*, *AtNRAMP3*, *AtNRAMP4*, and *AtNRAMP5*). *AtNRAMP3* and *AtNRAMP4* showed close homology with 4 NRAMPs from *G. max* (*GmNRAMP1a*, *GmNRAMP1b*, *GmNRAMP2a*, and *GmNRAMP2b*) and 4 genes from *A. hypodaea* (*AhNRAMP3.1*, *AhNRAMP3.2*, *AhNRAMP3.4*, and *AhNRAMP3.4*), while *StNRAMP4* was the outlier in this clade. AtNRAMP5 showed a close evolutionary relationship with *C. sinensis* and *S. tuberosum*. *AtNRAMP2* showed close homology with multiple NRAMPs from *Z. mays*, *T. aestivum*, *O. sativa*, *G. max*, and *A. hypogaea*.

Furthermore, we analyzed these *NRAMPs* to understand the relationship among all of these conserved metal ion-transporting orthologs. Circoletto tools were used to visualize the close relationship between *NRAMP* family members across these species. The Circoletto plot showed synteny of *AtNRAMP1* and *AtNRAMP6* with *GmNRAMP6a*. *AtNRAMP3* and *AtNRAMP4* were found to be closely linked with *GmNRAMP1b*. Lastly, *AtNRAMP5* and *AtNRAMP2* were closely associated with *AhNRAMP2.2* (Fig. 3).

Cis-acting elements regulatory analysis, Protein-Protein interaction, and 3D model prediction of *AtNRAMPs*

To understand the promoter regions of AtNRAMPs and how they are regulated, 1000 bp upstream sequences were downloaded and analyzed by the PlantCARE tool for cis-element analysis. After analysis, 45 cis elements were identified in these *AtNRAMPs* related to growth and development, hormone response, and abiotic stress response. All 45 cis-elements were visualized as Heatmap in TbTools (Fig. 5). The Maximum cis-regulatory elements found were related to the environment and stress-related effects. Among these most abundant

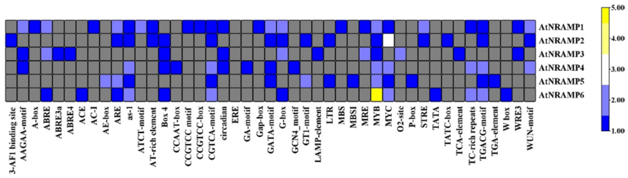

Fig. 5 Cis-Regulatory elements Heatmap made by TBtools. Yellow boxes indicate the maximum number of elements is one gene, and Navy blue shows the least count. Gray boxes indicate absence

elements were Box 4, found in all except *ATNRAMP1* and *AtNRAMP5*. Regulatory elements CGTCA-motif and TGACG-motif were abundant in all except *AtNRAMP3*. Other than that, GATA-motif and G-box were also abundant elements in the stress-responsive category. Among stress-responsive regulatory elements, MYB and MYC were found to be most present in all six *AtNRAMPs*, with the highest number in *AtNRAMP2* and *AtNRAMP6*. MYC and TGACG elements were found in 5 *AtNRAMPs* except *AtNRAMP3*. Additionally, abscisic acid response elements like ABRE were present in *AtNRAMP1*, *AtNRAMP3*, and *AtNRAMP6*, while not found in *AtNRAMP2* and *AtNRAMP5*, respectively.

Analysis of protein-protein interaction (PPI) networks revealed complex connections between important metal homeostasis proteins and AtNRAMP transporters. By interacting with many AtNRAMPs, Metal Tolerance Proteins (MTP9 and MTP11) stood out as key hubs, indicating a coordinated function in heavy metal detoxification. Furthermore, interactions with the oligopeptide transporter OPT7 and the calcium exchanger CAX4 suggest potential overlap between the metal ion detoxification, peptide transport, and calcium signaling pathways. The presence of the unidentified protein F2N1.2 points to an undiscovered involvement in metal homeostasis, whereas the significant interconnection across AtNRAMPs implies either functional redundancy or cooperative activities (Fig. 6).

Moreover, SWISS-MODEL was used to predict 3D models of AtNRAMPs (Fig. 7). *AtNRAMP1*, *AtNRAMP3*, *AtNRAMP4*, and *AtNRAMP6* well modeled against template Q9S9N8.1.A. *AtNRAMP2* and *AtNRAMP5* well modeled against template I1JTP4.1.A. The GMQE score ranged from 0.79 to 0.82, and Seq Identity was 100% for *AtNRAMP3*, *AtNRAMP4*, and *AtNRAMP6*, while for the remaining three, it ranged from 72 to 88%. These scores show that proteins were well modeled.

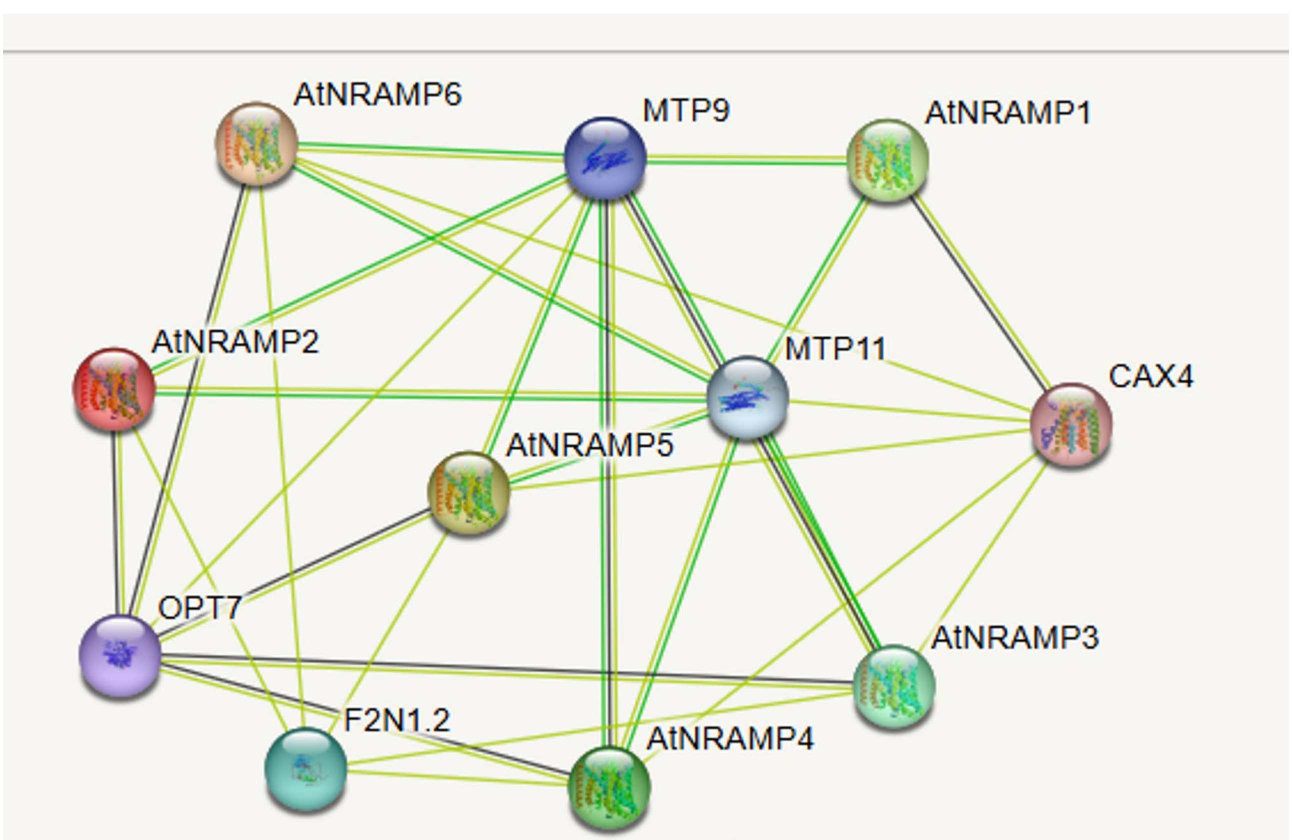

Fig. 6 Protein-protein interaction (PPI) network of *AtNRAMP* genes in *Arabidopsis thaliana* constructed using STRING. Thick lines indicate strong interactions, while thin lines represent weaker associations

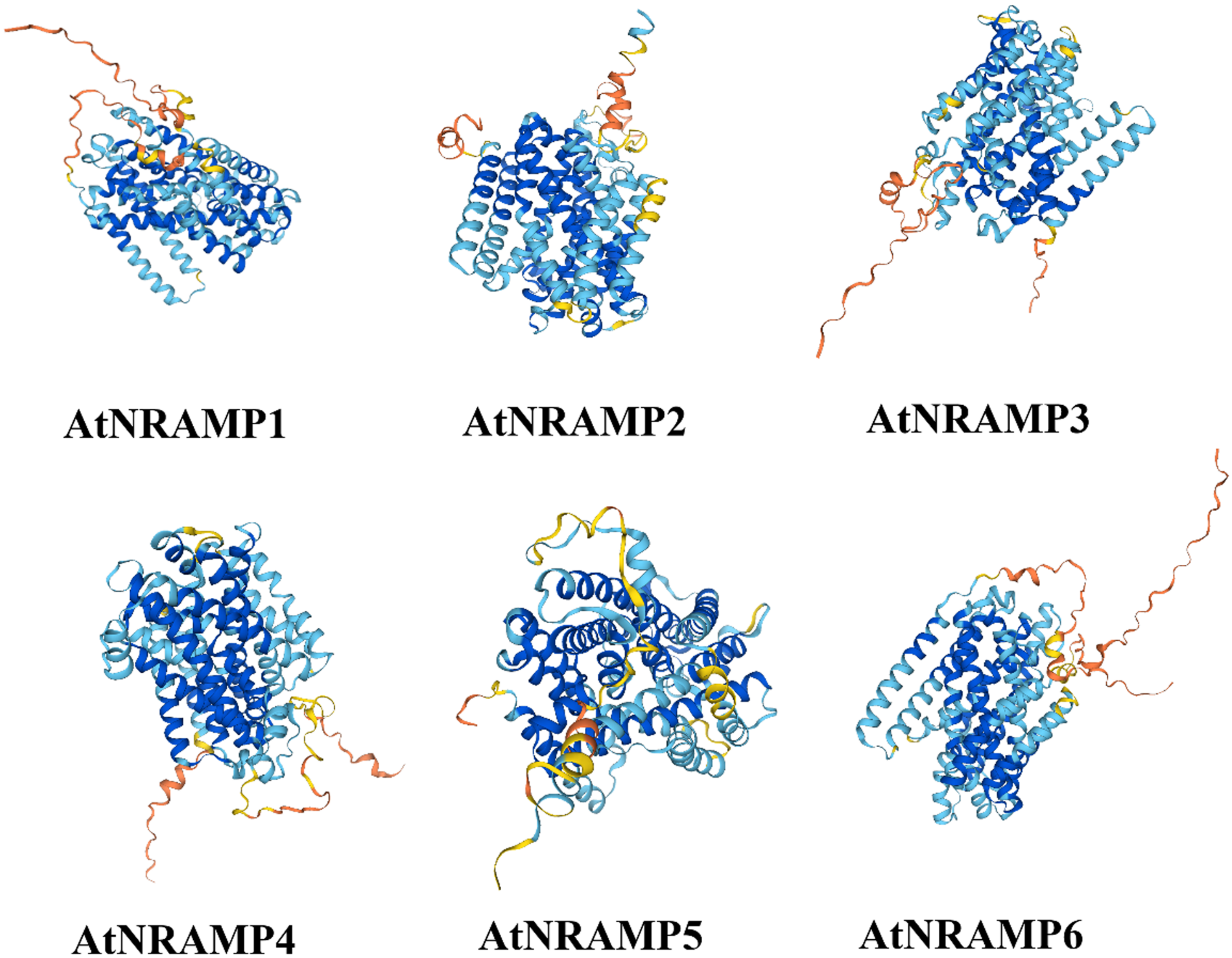

Fig. 7 Predicted 3D models of AtNRAMPs using SwissModel. The models are in rainbow colors from N to C terminus, with Blue being the most confident and orange the least

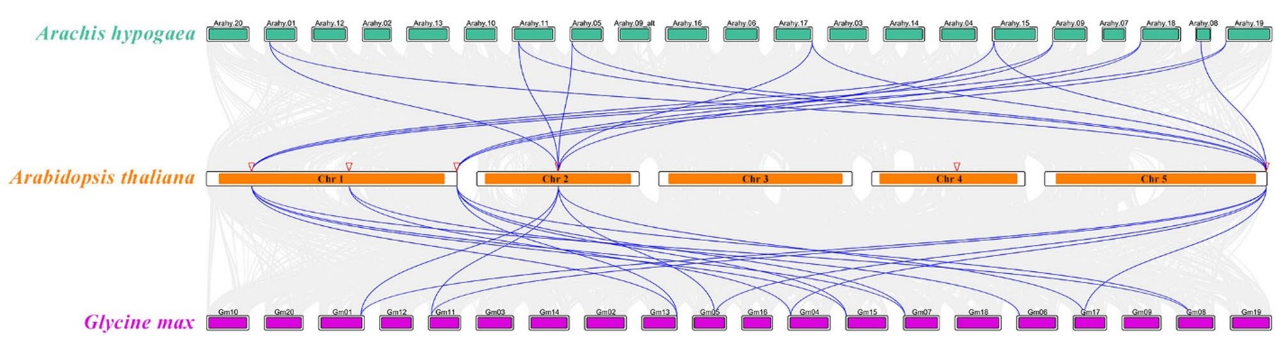

Fig. 8 Synteny analysis of *A.thaliana* with *G.max* and *A.hypogea*. Color bars (green, orange, and pink) show the chromosomes of *A.hypogea*, *A.thaliana*, and *G.max*. Blue lines indicate collinear genes across three genomes. Red inverted triangles show AtNRAMP1, AtNRAMP2, and AtNRAMP6 on Chr 1, AtNRAMP3 on Chr 2, AtNRAMP5 on Chr 5, and AtNRAMP4 on Chr 5. Gray lines show collinear blocks in all three genomes

Synteny analysis of *AtNRAMPs* with closely related species

To better understand the evolution of the *AtNRAMP* gene family, synteny analysis was performed on *A. thaliana*, *A. hypogea*, and *G. max*. This analysis revealed 17 gene pairs between *A. thaliana* and *A. hypogea* and 19 pairs between *A. thaliana* and *G. max* (Fig. 8). *AtNRAMP2* (Chr 1) had no collinear genes in *A. hypogea*, and similarly, *AtNRAMP5* (Chr 4) had no collinear genes in both *A. hypogea* and *G. max*. AtNRAMP Collinear genes in *A. hypogea* were mostly found on Chr 1, 11, 5, 17, 15, 9, 18, 8, and 19. In *G. max*, the collinear genes were found on Chr1, 11, 13, 5, 4, 15, 7, 6, 17, and 8. These high numbers of collinear genes show the close evolutionary relationship among these three species and the *AtNRAMP* gene family.

Expression profiling in different tissues and under different heavy metals stresses

The expression analysis of six *AtNRAMP* genes (*AtNRAMP1–AtNRAMP6*) in *Arabidopsis thaliana* revealed distinct tissue-specific patterns, with *AtNRAMP1* and *AtNRAMP5* showing the highest expression in roots, suggesting their primary role in root metal ion uptake and translocation, while *AtNRAMP3* and *AtNRAMP4* exhibited elevated expression in mature and young leaves, respectively, indicating potential involvement in leaf metal homeostasis or stress responses (Fig. 9). *AtNRAMP2* displayed moderate expression in flowers and stems, possibly implicating it in reproductive or structural tissue metal distribution, whereas *AtNRAMP6*

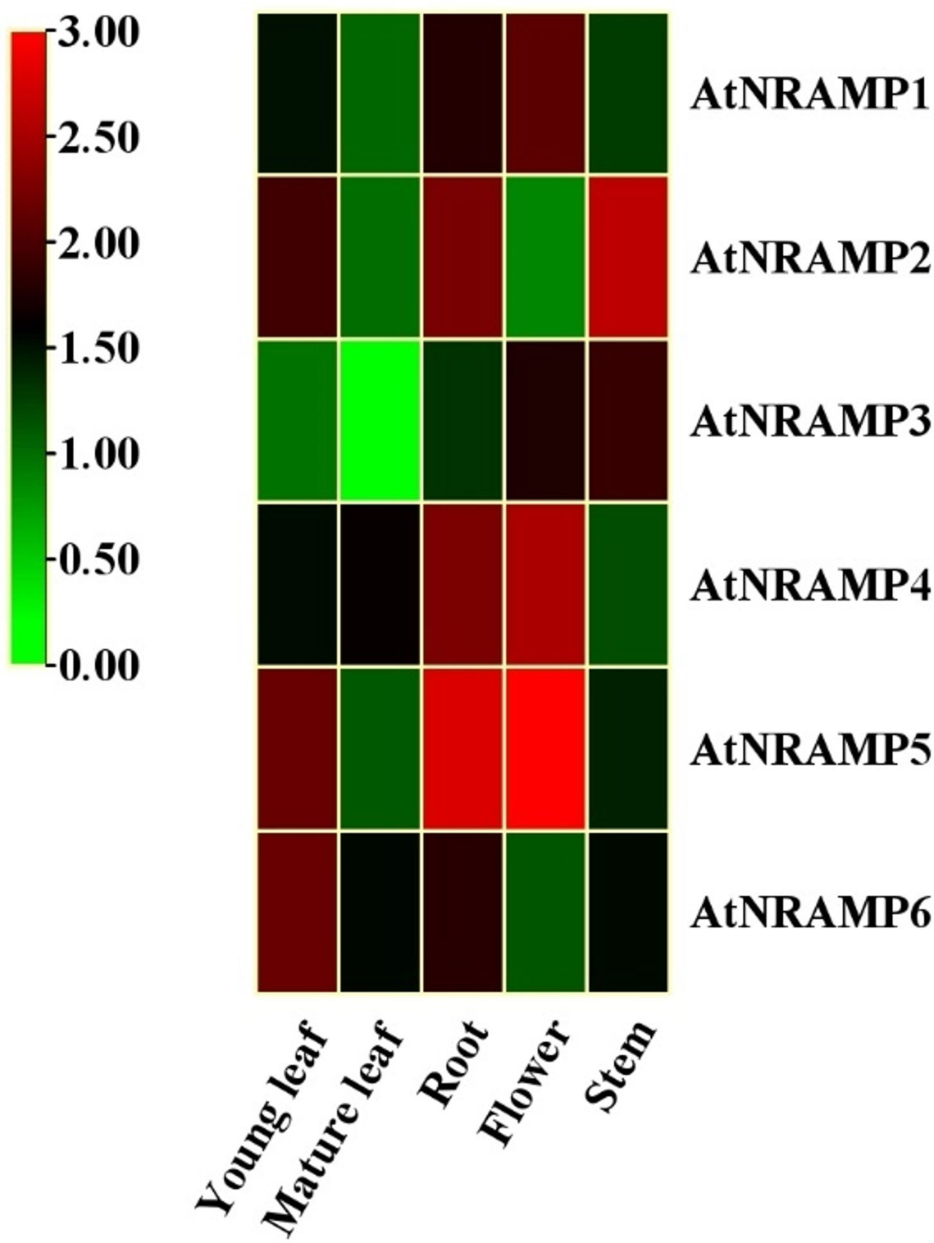

Fig. 9 Tissue-specific expression profiles of *AtNRAMP* genes in *Arabidopsis thaliana*. Relative mRNA levels of *AtNRAMP1–AtNRAMP6* across young leaf, mature leaf, root, flower, and stem tissues were determined by qRT-PCR. Expression values are normalized to actin gene (*AtACTIN2*) and data represent the mean ± standard deviation of three biological replicates

showed consistently low expression across all tissues, hinting at a specialized or conditionally regulated function (Fig. 9). These differential expression profiles highlight the functional diversification of the NRAMP family, with root-enriched members likely facilitating soil-metal acquisition and leaf-preferential members contributing to intracellular metal allocation, aligning with previous studies on NRAMP transporters in plant metal homeostasis [46]. The data, derived from three biological replicates, underscore the nuanced roles of these transporters in maintaining metal balance across different plant tissues, providing a foundation for further mechanistic studies into their tissue-specific regulatory networks and physiological contributions.

Under Cd treatment, *AtNRAMP3* and *AtNRAMP6* showed the strongest induction among all genes, with expression levels peaking significantly at later time points. *AtNRAMP1* also exhibited a noticeable upregulation, though to a lesser extent than *AtNRAMP3* and *AtNRAMP6*. Interestingly, *AtNRAMP2* and *AtNRAMP5* displayed a moderate increase in transcript levels, indicating their involvement in the Cd stress response. *AtNRAMP4*, although less responsive than others, showed a detectable increase, particularly at intermediate time points, suggesting that all six genes were transcriptionally responsive to cadmium stress to varying degrees (Fig. 10A). Under Cu exposure, *AtNRAMP1* was the most strongly induced gene, with high transcript accumulation observed over the time course. *AtNRAMP3*

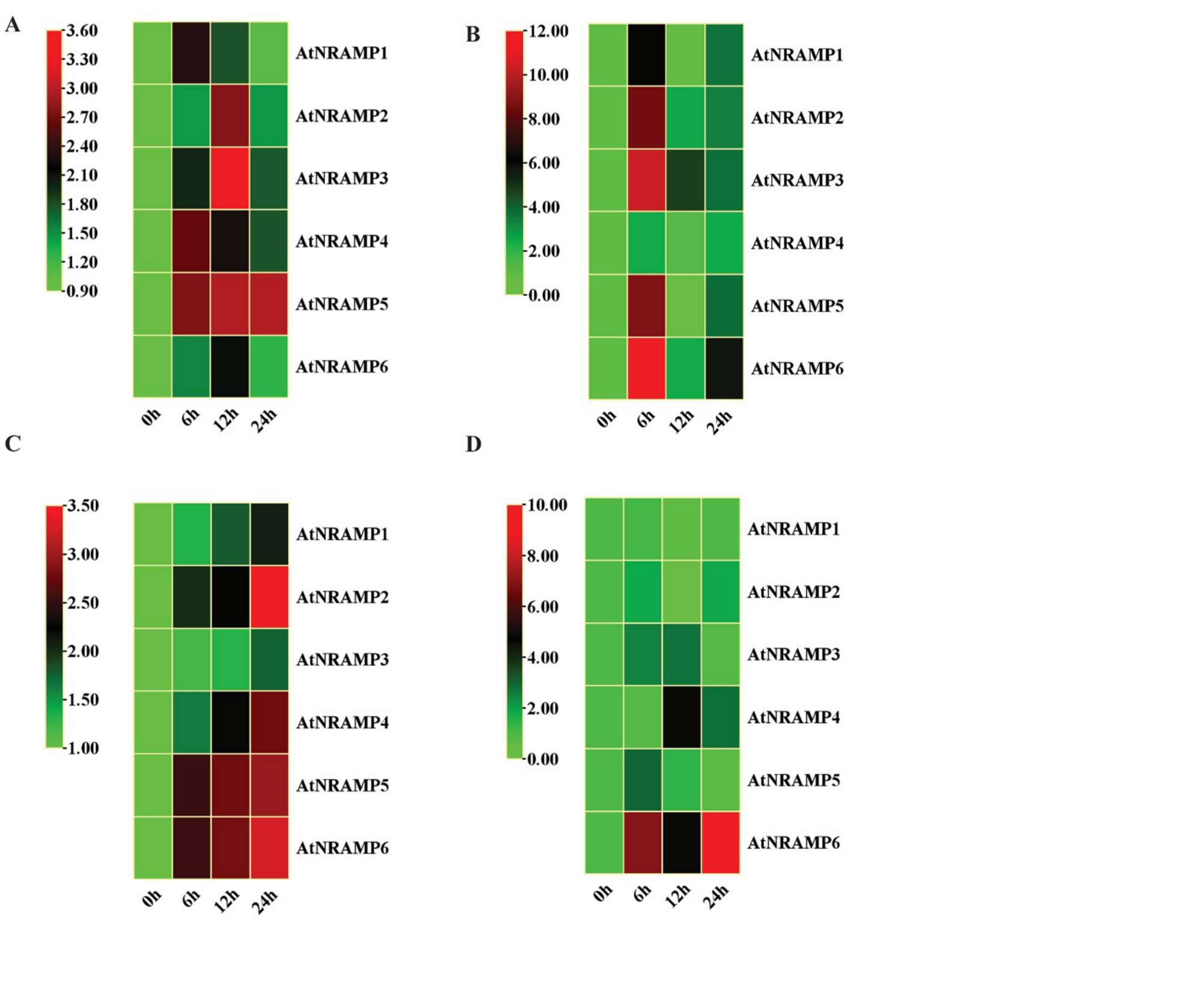

Fig. 10 Expression analysis of six *AtNRAMP* genes in response to different heavy metal stresses in *Arabidopsis thaliana*. Gene expression was analyzed by qRT-PCR after treatment with (A) cadmium (Cd), (B) copper (Cu), (C) lead (Pb), and (D) zinc (Zn) at indicated time points. Relative expression levels were normalized against an internal control gene, and expression at 0 h was used as the reference. Data represent the mean ± standard deviation of three biological replicates

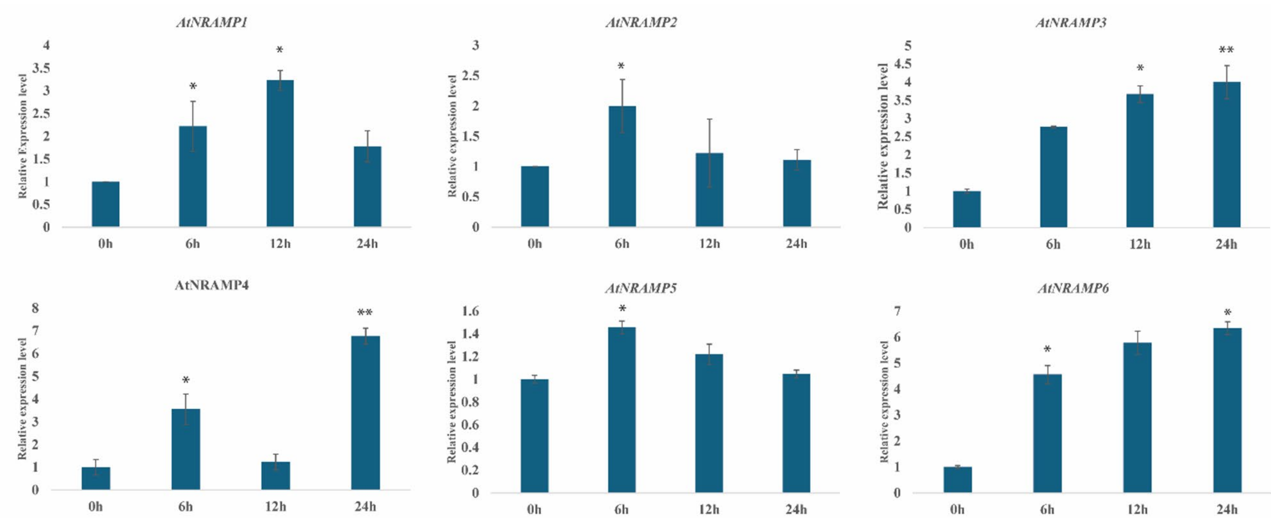

Fig. 11 Relative expression level of *AtNRAMP* genes in response to ABA stress. Transcription levels of six *AtNRAMPs* were examined under 5 µM ABA treatment for 0, 6, 12, and 24 h. Values are means ± SD of three replicates. * represent the significant differences at $p < 0.05$; ** represent $p < 0.01$ (Student's t-test)

and *AtNRAMP6* also showed upregulation, although the magnitude of their response was lower compared to Cd treatment. *AtNRAMP2* and *AtNRAMP4* exhibited mild but consistent expression, indicating potential secondary roles in Cu detoxification or transport. *AtNRAMP5* showed only a slight increase in transcript levels, suggesting a minimal involvement under copper stress (Fig. 10B).

In response to Pb stress, *AtNRAMP2* and *AtNRAMP5* displayed moderate expression levels, with noticeable transcript accumulation, particularly at later stages, indicating that these two genes are more Pb-responsive than previously assumed. *AtNRAMP3* and *AtNRAMP6* showed relatively lower induction compared to their response under Cd and Cu stress. *AtNRAMP1* and *AtNRAMP4* exhibited weak responses, with only slight changes in expression, suggesting a limited role in lead stress adaptation for these two members (Fig. 10C). During Zn treatment, *AtNRAMP3* and *AtNRAMP6* again showed prominent upregulation, consistent with their behavior under other metal stresses. *AtNRAMP1* and *AtNRAMP4* displayed moderate increases in transcript levels, while *AtNRAMP2* and *AtNRAMP5* also showed detectable expression, particularly at early time points, although the overall response was lower than that of *AtNRAMP3* and *AtNRAMP6*. These results suggest that all six genes are transcriptionally responsive to zinc, with varying degrees of induction (Fig. 10D).

Expression profiling in response to phytohormones abscisic acid (ABA)

In response to abscisic acid (ABA) treatment, the expression of the six *AtNRAMP* genes displayed differential regulation patterns that correspond closely with the presence of ABRE (ABA-responsive) cis-acting regulatory elements in their promoter regions. *AtNRAMP1*, *AtNRAMP3*, and *AtNRAMP6*, which contain ABRE elements, showed significant induction in expression levels, suggesting that these genes are transcriptionally responsive to ABA signaling (Fig. 11). Among them, *AtNRAMP3* exhibited the highest expression increase, highlighting its potential role as a key mediator in ABA-dependent stress responses (Fig. 11). In contrast, *AtNRAMP2* and *AtNRAMP5*, which lack ABRE motifs, showed little to no expression change, supporting the idea that their promoters are not directly responsive to ABA. *AtNRAMP4* showed a moderate induction despite the unclear presence of ABRE elements, implying the possible involvement of alternative ABA-responsive elements or indirect regulatory mechanisms (Fig. 11). These observations suggest that *AtNRAMP1*, *AtNRAMP3*, and *AtNRAMP6* are likely important components of ABA-mediated stress adaptation pathways, potentially contributing to metal ion transport and homeostasis under abiotic stress conditions.

Discussion

NRAMPs are a prominent gene family that exists in a wide range of bacteria, animals, and plants that play a significant role in the absorption and transport of heavy metals such as Iron (Fe), Copper (Cu), Zinc (Zn), Lead (Pb), Cadmium (Cd), and Manganese [10]. The *NRAMPs* also show a crucial impact in response to heavy metal stress. Till now, the NRAMP gene family has been reported in a number of plant species [47, 48], but there is an abundance of research that has been carried out on the genome-wide study of the *NRAMP* gene family in

Arabidopsis thaliana. We retrieved 77 NRAMP members from 8 species, including *Arabidopsis thaliana*, *Solanum tuberosum*, *Triticum aestivum*, *Zea mays*, *Glycine max*, *Oryza sativa*, *Arachis hypogaea*, and *Camellia sinensis*. In this study, the Arabidopsis genome was comprehensively searched for *NRAMP* genes for genome-wide identification, which resulted in the identification of 6 putative *AtNRAMP* genes based on the conserved domain of identified AtNRAMP family proteins, and assessed their functional and structural properties. They exhibited differential expression under heavy metal stress, suggesting their critical roles in metal homeostasis. Subsequently, they were designated as *AtNRAMP1* through *AtNRAMP6*.

The physicochemical characterization of the six *Arabidopsis thaliana* NRAMP proteins revealed detailed profiles, including molecular weight, amino acid length, isoelectric point (pI), and hydrophobicity. The *AtNRAMP* proteins exhibited pI values ranging from 4.99 to 8.86, indicating their ability to acquire different net charges depending on environmental pH conditions. All *AtNRAMPs* showed positive GRAVY (Grand Average of Hydropathicity) values, reflecting high hydrophobicity, a feature essential for stable membrane association and functionality in cellular systems [49]. Consistent with previous findings, each *AtNRAMP* protein was predicted to contain 12 transmembrane domains (TMDs), underlining the structural basis of their role as membrane-bound transporters [50]. Recently, high-resolution structural and computational studies have clarified the exact metal transportation and binding pathways in Nramp transporters, which unveiled molecular explanations for the conserved substrate specificity across species [35, 51].

The duplication analysis provided important insights into the evolutionary dynamics of the *AtNRAMP* gene family. A total of nine duplicated gene pairs were identified, comprising one tandem duplication and eight segmental duplications, reflecting the complexity of gene family expansion in *Arabidopsis*. The observed ratio of non-synonymous (Ka) to synonymous (Ks) substitutions indicated that non-synonymous changes occurred less frequently, suggesting purifying selection to preserve the original gene functions. These findings align with previous reports showing that synonymous mutations are typically neutral unless they occur in regulatory or structurally sensitive regions [52, 53]. Furthermore, the analysis supports the concept that synonymous substitutions can influence gene expression and stability, reinforcing their potential regulatory significance under evolutionary pressure [54]. Consistent studies in poplar and foxtail millet have indeed confirmed the importance of purifying selection on the original NRAMP gene function after duplication, with adaptive structural changes serving environmental stress response [38, 55, 56].

Phylogenetic analysis of the *AtNRAMP* gene family corroborated earlier findings on their evolutionary divergence and functional specialization, revealing their classification into two distinct clades, as previously reported in *Arabidopsis* [57]. Clade 1, comprising *AtNRAMP2*, *AtNRAMP3*, and *AtNRAMP4*, was characterized by a reduced number of introns, a feature commonly associated with functional streamlining. In contrast, Clade 2 members, including *AtNRAMP1* and *AtNRAMP6*, exhibited more complex exon–intron structures and the exclusive presence of motif 10, which may reflect clade-specific adaptations to stress, as also observed by [29]. Variation in gene structure and motif distribution underscores adaptive evolution among NRAMP paralogs in different plant species, facilitating a broad range of metal transport functions tailored to diverse physiological requirements [48]. The presence of highly conserved motifs such as motifs 1, 2, 3, 5, and 6 across all *AtNRAMPs* highlights their essential and conserved roles in divalent metal ion transport [58]. Additionally, the conservation of transmembrane domains affirms their shared function as membrane-localized metal transporters, and the presence of regulatory elements in untranslated regions (UTRs) suggests post-transcriptional control in response to environmental stress [34]. Although further functional validation is needed to clarify the specific role of unique motifs such as motif 10 in stress tolerance, these findings underscore the structural complexity and functional versatility of the *AtNRAMP* gene family in metal transport and stress adaptation.

We conducted a comprehensive phylogenetic analysis of NRAMP (Natural Resistance-Associated Macrophage Protein) genes across multiple plant species to explore their evolutionary relationships and functional diversification. The resulting phylogenetic tree revealed three distinct clades, each with unique gene compositions and implied functional specializations. Clade 1 consisted predominantly of monocot species such as *Triticum aestivum* (*TaNRAMPs*) and *Oryza sativa* (*OsNRAMPs*), along with representatives from *Arachis hypogaea* (*AhNRAMPs*), *Zea mays* (*ZmNRAMPs*), *Camellia sinensis* (*CsNRAMPs*), and *Solanum tuberosum* (*StNRAMPs*) [32, 33, 39]. The absence of *Arabidopsis thaliana* NRAMPs (AtNRAMPs) from this clade suggests a divergent evolutionary pathway between monocots and dicots [34, 35]. The clustering of *TaNRAMPs* and *OsNRAMPs* within Clade 1 implies a conserved role in metal transport under abiotic stress conditions specific to monocots. Clade 2 included *AtNRAMP1* and *AtNRAMP6*, grouped alongside *OsNRAMP3*, *AmNRAMP4*, and several wheat NRAMPs, reflecting their conserved involvement in metal homeostasis and stress responses across both monocots and dicots [29, 59]. The composition of this clade also suggests evolutionary adaptations in response to Lineage-specific physiological demands, particularly in legumes. Clade 3 contained *AtNRAMP2*, *AtNRAMP3*, *AtNRAMP4*, and *AtNRAMP5*, along with NRAMP genes from species such as *Glycine max*, *A. hypogaea*, and *S. tuberosum*. This clustering highlights conserved roles in legumes, with functional divergence observed among NRAMPs in *S. tuberosum*. Consistent with previous studies, *AtNRAMP3* and *AtNRAMP4* showed close clustering with legume NRAMPs, supporting their established roles in divalent metal transport and stress tolerance. *AtNRAMP2*, which exhibited broad homology across diverse species, appears to play a central role in mediating responses to abiotic stress. Overall, the phylogenetic structure reflects both evolutionary conservation and divergence of the NRAMP gene family [47]. Clade 1 shows monocot-specific specialization, whereas Clades 2 and 3 display mixed compositions, representing both conserved functions and species-specific adaptations in metal transport and stress resilience [35].

Synteny analysis using Circoletto revealed that NRAMP family members are evolutionarily conserved and functionally aligned across a wide range of plant species, particularly in their roles as metal ion transporters. In *Arabidopsis thaliana* and *Glycine max*, *AtNRAMP1* and *AtNRAMP6* displayed strong syntenic relationships with *GmNRAMP6a*, indicating conserved functions likely related to metal ion transport and stress tolerance mechanisms. Similarly, *AtNRAMP3* and *AtNRAMP4* showed high collinearity with *GmNRAMP1b*, reinforcing their involvement in shared pathways governing metal homeostasis and detoxification, consistent with findings in leguminous species. Furthermore, *AtNRAMP2* and *AtNRAMP5* exhibited synteny with *AhNRAMP2.2* from *Arachis hypogaea*, suggesting conserved roles in ion uptake and transport between these distantly related species. This cross-species conservation aligns with the known functions of NRAMP proteins in maintaining metal ion availability, especially under variable environmental conditions [33, 47]. In addition to regulating homeostasis, such syntenic relationships underscore the role of NRAMPs in enabling plants to adapt to abiotic stresses such as metal toxicity and nutrient deficiencies [60]. Collectively, these findings highlight the evolutionary significance of *NRAMP* genes as integral and conserved components of the plant metal ion transport network.

Cis-regulatory element analysis identified 45 distinct regulatory motifs within the 1000 bp upstream promoter regions of *AtNRAMP* genes, revealing their involvement in diverse biological processes, including growth, development, hormonal signaling, and environmental stress responses. Stress-related cis-elements were widely distributed among the gene family, with MYB-binding sites notably present in all *AtNRAMP* promoters, most prominently in *AtNRAMP6*, highlighting their potential regulation by MYB transcription factors known to modulate metal homeostasis and abiotic stress tolerance [61].

All *AtNRAMP* genes, except *AtNRAMP3*, contained key jasmonic acid-responsive motifs, namely the CGTCA-motif and TGACG-motif, suggesting a role in defense and stress adaptation pathways mediated by jasmonate signaling [62]. MYC elements, associated with abscisic acid (ABA) signaling and responses to drought and salinity stress, were present in five *AtNRAMPs*, again excluding *AtNRAMP3*, indicating gene-specific regulatory divergence [63]. Light- and stress-responsive elements such as the G-box and GATA motifs were also identified across the family, suggesting coordinated regulation in response to oxidative and nutrient-related stress cues. Notably, *AtNRAMP3* lacked both MYC and TGACG motifs, while Box 4, a light-responsive element, was absent in *AtNRAMP1* and *AtNRAMP5*, reflecting distinct regulatory architectures among gene members. These findings, aligned with previous reports of functional divergence in NRAMP genes [59], emphasize the complex transcriptional control underlying *AtNRAMP* gene function and underscore the need for further functional studies to elucidate the roles of these cis-elements in metal stress responses. More recent comprehensive reviews further highlight that phytohormones like abscisic acid, jasmonic acid, and salicylic acid are involved in the modulation of NRAMP gene expression during heavy metal stress, indicating complex multi-hormonal (crosstalk) adaptation to stress [22, 24].

Protein–protein interaction (PPI) network analysis in plants revealed critical insights into the regulatory framework of metal homeostasis, underscoring the central role of membrane-associated proteins and their interactions with metal ions and lipids. Within this network, *AtNRAMP* transporters were closely associated with key hub proteins such as *MTP9* and *MTP11*, which are known to facilitate heavy metal detoxification. These interactions suggest that *MTPs* serve as coordination centers in managing cellular metal ion balance. Consistent with previous findings, *NRAMP* proteins were confirmed to participate in the transport and regulation of essential divalent cations, including iron (Fe^{2+}), manganese (Mn^{2+}), and cadmium (Cd^{2+}), which are vital for maintaining cellular homeostasis under metal stress conditions [64]. The network also revealed functional links between *AtNRAMPs* and peptide transporters such as *OPT7* and *CAX4*, suggesting an integrated mechanism of metal detoxification that involves broader signaling and transport systems. Notably, the identification of a previously uncharacterized protein, *F2N1.2*, within the interaction network points to the potential presence of novel components involved in metal ion regulation. While the function of this protein remains to be elucidated, its inclusion in the network highlights the need for further investigation to determine its role in metal homeostasis [46, 65]. Collectively, the interaction network emphasizes the complex and coordinated nature of plant responses to heavy metal stress, with *NRAMP* proteins functioning as integral components of this regulatory system. Recently, functional characterizations have broadened applications for NRAMP transporters to plant pathogen defense and salt stress tolerance, important for adaptive responses during changing environments [35, 39].

The three-dimensional structures of *AtNRAMP* proteins were predicted using the SWISS-MODEL platform. *AtNRAMP1*, *AtNRAMP3*, *AtNRAMP4*, and *AtNRAMP6* were modeled using the Q9S9N8.1.A template, while *AtNRAMP2* and *AtNRAMP5* were aligned with the I1JTP4.1.A template [66, 67]. The generated models exhibited high reliability, with GMQE (Global Model Quality Estimation) scores ranging from 0.79 to 0.82. Sequence identity was 100% for *AtNRAMP3*, *AtNRAMP4*, and *AtNRAMP6*, and ranged from 72% to 88% for *AtNRAMP1*, *AtNRAMP2*, and *AtNRAMP5*. Model accuracy was visually represented by a color gradient from blue (high confidence) to orange (low confidence), consistent with prior modeling standards, further supporting the structural validity of the predicted models.

The evolutionary relationships of *NRAMP* genes were further explored through synteny analysis between *Arabidopsis thaliana* and its closely related leguminous species, *Arachis hypogaea* and *Glycine max* [34]. Comparative genomic analysis revealed 17 collinear gene pairs between *A. thaliana* and *A. hypogaea*, and 19 collinear pairs between *A. thaliana* and *G. max*, indicating a high degree of evolutionary conservation across these species. However, *AtNRAMP2* (located on chromosome 1) and *AtNRAMP5* (on chromosome 4) did not show syntenic counterparts in either species. This lack of collinearity may reflect gene loss or divergence events that occurred during lineage separation, consistent with models of gene family evolution proposed in [68]. In *A. hypogaea*, conserved syntenic regions were primarily found on chromosomes 1, 5, 8, 9, 11, 15, 17, 18, and 19, while in *G. max*, syntenic blocks were identified on chromosomes 1, 4, 5, 6, 7, 8, 11, 13, 15, and 17. These results align with earlier reports highlighting the conserved nature of NRAMP genes across plant lineages [69], further reinforcing their essential roles in metal ion transport and stress response mechanisms. Additionally, the recent comparative genomic analyses indicate that the syntenic conservation among NRAMP genes supports integrated regulation and maintenance of metal homeostasis in legume and dicot plants alongside lineage-specific expansions and contractions that produce adaptive functional diversification [34, 36]. Such genome rearrangements likely reflect evolutionary pressures such as soil heterogeneity and heavy metal uptake demands that select for divergence of the NRAMP family to better adapt to stress. The observed absence of synteny for certain *AtNRAMP* members suggests lineage-specific evolutionary divergence, potentially driven by adaptive pressures or genome rearrangements unique to each species.

Moreover, individual genes exhibited distinct expression patterns in response to various heavy metal stresses and hormones, with tissue-specific differences in their regulatory behavior. This study identified five *NRAMP* family genes in *Arabidopsis* and examined their sequences and gene structures in detail. Expression profiling using qRT-PCR revealed how these genes respond differentially to various heavy metal treatments, offering valuable genetic insights for understanding the mechanisms of heavy metal uptake and accumulation in Arabidopsis.

Conclusion

In this study, we systematically identified and characterized the *NRAMP* gene family in *Arabidopsis thaliana*, providing comprehensive insights into their roles in heavy metal stress responses. Six *AtNRAMP* genes were analyzed in detail with respect to their physicochemical properties, evolutionary relationships, gene duplication events, and regulatory features. The conserved presence of 12 transmembrane domains suggests structural and functional stability, while variations in gene structure and motif composition indicate potential functional specialization. Phylogenetic and synteny analyses revealed evolutionary divergence and regulatory complexity across plant species. Notably, predicted interactions between *AtNRAMP* proteins and other metal transporters offer valuable leads for future functional studies. Three-dimensional structural modeling further supports their putative roles in metal ion transport. Expression profiling under various heavy metal and hormonal treatments revealed differential gene responses, underscoring functional divergence within the gene family. Continued investigation into their expression under specific heavy metal stresses will be critical to fully elucidate their roles in metal homeostasis. Collectively, these findings enhance our understanding of metal transport mechanisms in plants and may contribute to the development of strategies for improving crop resilience in metal-contaminated environments.

Limitations of the study

Although this study provides a comprehensive characterization of the *NRAMP* gene family in *Arabidopsis thaliana*, several limitations should be acknowledged. The findings are largely based on computational predictions and gene expression profiling. While we performed expression analyses under selected heavy metal and hormonal treatments, further experimental work is required to validate these responses at the protein level and under broader stress conditions. In particular, functional assays such as metal transport activity, subcellular localization, and protein–protein interaction studies were not conducted, leaving uncertainties about the precise roles of individual *NRAMPs*. Moreover, the three-dimensional structural models generated were predictive and require validation through high-resolution structural approaches. Addressing these limitations in future research will be essential to fully unravel the roles of *NRAMPs* in metal homeostasis and their potential applications in developing stress-resilient crops.

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

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Authors' contributions

Muhammad Arif: Conceptualization, Methodology, Software, Data Curation, Writing-original draft, Writing-review & editing. Hina Abbas: Methodology, Software, Writing-review & editing. Noman Mahmood: Methodology, Data Curation, Software, Writing-review & editing. Muhammad Uzair: Methodology, Software, Writing-review & editing. Muhammad Aamir Manzoor: Methodology, Data curation, Writing-review & editing. Shahbaz Atta Tung: Writing- review & editing. Yao Xin: Methodology, Software, Writing-review & editing. Ruhong Xu: Conceptualization, Supervision, Validation, funding acquisition, Writing-review & editing. Luhua Li: Conceptualization, Supervision, Project administration, funding acquisition, Writing-review & editing.

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Data availability

Data will be made available on request from Corresponding author.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. Riyazuddin R, Nisha N, Ejaz B, Khan MIR, Kumar M, Ramteke PW, Gupta R. A comprehensive review on the heavy metal toxicity and sequestration in plants. Biomolecules. 2021;12(1):43.
2. Arif M, Khalid F, Abbas H, Maqsood S, Khan SU, Adnan M, Ghorbani A. Climate change impact on soil systems and agriculture resilience. Soils and sustainable agriculture: interplay of soil, plant, water and environmental systems for sustainable agriculture. Springer; 2025. pp. 381–98. https://doi.org/10.1007/978-3-031-91114-9_15.
3. Ghori N-H, Ghori T, Hayat M, Imadi S, Gul A, Altay V, et al. Heavy metal stress and responses in plants. Int J Environ Sci Technol. 2019;16(3):1807–28.
4. Rashid A, Schutte BJ, Ulery A, Deyholos MK, Sanogo S, Lehnhoff EA, Beck L. Heavy metal contamination in agricultural soil: environmental pollutants affecting crop health. Agronomy. 2023;13(6):1521.
5. Charagh S, Hui S, Wang J, Raza A, Zhou L, Xu B, et al. Unveiling innovative approaches to mitigate metals/metalloids toxicity for sustainable agriculture. Physiol Plant. 2024;176(2):e14226.
6. Sterckeman T, Thomine S. Mechanisms of cadmium accumulation in plants. CRC Crit Rev Plant Sci. 2020;39(4):322–59.
7. Tan Z, Li J, Guan J, Wang C, Zhang Z, Shi G. Genome-wide identification and expression analysis reveals roles of the NRAMP gene family in iron/cadmium interactions in peanut. Int J Mol Sci. 2023;24(2):1713.
8. Arif M, Ren M, Xu R, Adnan M, Fahad S, Chen MX, et al. Biotechnological and multiomics strategies for understanding and controlling cadmium stress in wheat. Environmental Chemistry and Ecotoxicology. 2025. https://doi.org/10.1016/j.enceco.2025.07.020.
9. Palansooriya KN, Shaheen SM, Chen SS, Tsang DC, Hashimoto Y, Hou D, et al. Soil amendments for immobilization of potentially toxic elements in contaminated soils: a critical review. Environ Int. 2020;134:105046.
10. El Rasafi T, Oukarroum A, Haddioui A, Song H, Kwon EE, Bolan N, et al. Cadmium stress in plants: a critical review of the effects, mechanisms, and tolerance strategies. Crit Rev Environ Sci Technol. 2022;52(5):675–726.
11. Kim DY, Bovet L, Maeshima M, Martinoia E, Lee Y. The ABC transporter AtPDR8 is a cadmium extrusion pump conferring heavy metal resistance. Plant J. 2007;50(2):207–18.
12. Mendoza-Cózatl DG, Jobe TO, Hauser F, Schroeder JI. Long-distance transport, vacuolar sequestration, tolerance, and transcriptional responses induced by cadmium and arsenic. Curr Opin Plant Biol. 2011;14(5):554–62.
13. Memon A, Meraklı N. Comparative structural analysis of heavy metal ATPases in Arabidopsis thaliana, Arabidopsis halleri, brassica rapa, and brassica juncea. Turk J Agric. 2022;10:2988–95.
14. Li S, Yu J, Zhu M, Zhao F, Luan S. Cadmium impairs ion homeostasis by altering K + and Ca2 + channel activities in rice root hair cells. Plant Cell Environ. 2012;35(11):1998–2013.
15. Yuan HM, Huang X. Inhibition of root meristem growth by cadmium involves nitric oxide-mediated repression of auxin accumulation and signalling in Arabidopsis. Plant Cell Environ. 2016;39(1):120–35.
16. Jamla M, Khare T, Joshi S, Patil S, Penna S, Kumar V. Omics approaches for understanding heavy metal responses and tolerance in plants. Curr Plant Biol. 2021;27:100213.
17. Molina-Moya E, Rodríguez-González A, Peláez-Vico MA, Sandalio LM, Romero-Puertas MC. Peroxisomal-dependent signalling and dynamics modulate plant stress responses: reactive oxygen and nitrogen species as key molecules. J Exp Bot. 2025. https://doi.org/10.1093/jxb/eraf072.
18. Bücker-Neto L, Paiva ALS, Machado RD, Arenhart RA, Margis-Pinheiro M. Interactions between plant hormones and heavy metals responses. Genet Mol Biol. 2017;40(1 suppl 1):373–86.
19. Sytar O, Ghosh S, Malinska H, Zivcak M, Brestic M. Physiological and molecular mechanisms of metal accumulation in hyperaccumulator plants. Physiol Plant. 2021;173(1):148–66.
20. Zhang Y, Yang C, Liu S, Xie Z, Chang H, Wu T. Phytohormones-mediated strategies for mitigation of heavy metals toxicity in plants focused on sustainable production. Plant Cell Rep. 2024;43(4):99.
21. Thakur A, Singh A, Tandon A, Sharma V. Insights into the molecular mechanisms of uptake, phytohormone interactions and stress alleviation by silicon: a beneficial but non-essential nutrient for plants. Plant Growth Regul. 2023;101(1):1–13.
22. Rahman SU, Li Y, Hussain S, Hussain B, Khan W-u-D, Riaz L, et al. Role of phytohormones in heavy metal tolerance in plants: a review. Ecol Indic. 2023;146:109844.

23. Singh S, Parihar P, Singh R, Singh VP, Prasad SM. Heavy metal tolerance in plants: role of transcriptomics, proteomics, metabolomics, and ionomics. Front Plant Sci. 2016;6:1143.
24. Samanta S, Roychoudhury A. Molecular crosstalk of jasmonate with major phytohormones and plant growth regulators during diverse stress responses. J Plant Growth Regul. 2025;44(1):62–88.
25. Cellier M, Prive G, Belouchi A, Kwan T, Rodrigues V, Chia W, et al. Nramp defines a family of membrane proteins. Proc Natl Acad Sci U S A. 1995;92(22):10089–93.
26. Pinner E, Gruenheid S, Raymond M, Gros P. Functional complementation of the yeast divalent cation transporter family SMF by NRAMP2, a member of the mammalian natural resistance-associated macrophage protein family. J Biol Chem. 1997;272(46):28933–8.
27. Chen K, Jiang X, Zhu Z. Advances in the study of plant Nramp family involved in metal ion absorption and distribution. Plant Physiol J. 2020;56(3):345–55.
28. Curie C, Alonso JM, Jean ML, Ecker JR, Briat J-F. Involvement of NRAMP1 from *Arabidopsis thaliana* in iron transport. Biochem J. 2000;347(3):749–55.
29. Cailliatte R, Schikora A, Briat J-F, Mari S, Curie C. High-affinity manganese uptake by the metal transporter NRAMP1 is essential for Arabidopsis growth in low manganese conditions. Plant Cell. 2010;22(3):904–17.
30. Yue X, Song J, Fang B, Wang L, Zou J, Su N, et al. BcNRAMP1 promotes the absorption of cadmium and manganese in Arabidopsis. Chemosphere. 2021;283:131113.
31. Li J, Wang Y, Zheng L, Li Y, Zhou X, Li J, Gu D, Xu E, Lu Y, Chen X. The intracellular transporter AtNRAMP6 is involved in Fe homeostasis in Arabidopsis. Front Plant Sci. 2019;10:1124.
32. Qin L, Han P, Chen L, Walk TC, Li Y, Hu X, et al. Genome-wide identification and expression analysis of NRAMP family genes in soybean (*Glycine max* L). Front Plant Sci. 2017;8:1436.
33. Tian W, He G, Qin L, Li D, Meng L, Huang Y, et al. Genome-wide analysis of the NRAMP gene family in potato (*Solanum tuberosum*): identification, expression analysis and response to five heavy metals stress. Ecotoxicol Environ Saf. 2021;208:111661.
34. Zhao Y, Xie Q, Yang Q, Cui J, Tan W, Zhang D, Xiang J, Deng L, Guo Y, Li M. Genome-wide identification and evolutionary analysis of the NRAMP gene family in the AC genomes of brassica species. BMC Plant Biol. 2024;24(1):311.
35. Ma J, Li S, Pehlivan N, Zaman S, Anwar A. Genome-wide identification and evolutionary analysis of NRAMP family genes in tomato (*Solanum lycopersicum* L.) under cadmium and salt stress. BMC Genomics. 2025;26(1):759.
36. Li X, Cai H, Tian P, Zhao K, Wan J, Yin T, Zhang H, Liu X. Genome-wide identification of the NRAMP gene family in Kiwifruit and the response of AcNRAMP7 to cadmium stress. 2025.
37. Li H, Li Y, Yang F, Xiaolin L, Yifan D, Wang N, et al. Unraveling the NRAMP gene family: *Aegilops tauschii*'s prominent barrier against metal stress. Agronomy. 2025;15(8):1919.
38. Hu X, Li X, Zhu B, Gu L, Zeng T, Yu F, et al. Genome-wide identification of natural resistance-associated macrophage protein (NRAMP) and expression analysis under heavy metal stress in *sorghum bicolor* L. Plants. 2025;14(17):2660.
39. Yu Y, Rong K, Sui X, Zhang L, Zhang M, Hu H, et al. Analysis of NRAMP genes in the triticeae reveals that TaNRAMP5 positively regulates cadmium (Cd) tolerance in wheat (*Triticum aestivum*). Plant Physiol Biochem. 2025;219:109321.
40. Li J, Duan Y, Han Z, Shang X, Zhang K, Zou Z, et al. Genome-wide identification and expression analysis of the NRAMP family genes in tea plant (*Camellia sinensis*). Plants. 2021;10(6):1055.
41. Lescot M, Déhais P, Thijs G, Marchal K, Moreau Y, Van de Peer Y, Rouzé P, Rombauts S. PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for in Silico analysis of promoter sequences. Nucleic Acids Res. 2002;30(1):325–7.
42. Choudhuri S. Bioinformatics for beginners: genes, genomes, molecular evolution, databases and analytical tools. Elsevier; 2014.
43. Arif M, Li Z, Luo Q, Li L, Shen Y, Men S. The BAG2 and BAG6 genes are involved in multiple abiotic stress tolerances in *Arabidopsis Thaliana*. Int J Mol Sci. 2021;22(11):5856.
44. Han M, Feng J, Zhai Y, Si Y, Liu X, Han Y, et al. Promoted metabolic remolding by overexpression of AspAT9 ameliorates cadmium toxicity in Arabidopsis. J Hazard Mater. 2025;483:136688.
45. Yu CS, Lin CJ, Hwang JK. Predicting subcellular localization of proteins for gram-negative bacteria by support vector machines based on n-peptide compositions. Protein Sci. 2004;13(5):1402–6.
46. Bashir K, Rasheed S, Kobayashi T, Seki M, Nishizawa NK. Regulating subcellular metal homeostasis: the key to crop improvement. Front Plant Sci. 2016;7:1192.
47. Ishida JK, Caldas DG, Oliveira LR, Frederici GC, Leite LMP, Mui TS. Genome-wide characterization of the NRAMP gene family in phaseolus vulgaris provides insights into functional implications during common bean development. Genet Mol Biol. 2018;41(4):820–33.
48. Meng JG, Zhang XD, Tan SK, Zhao KX, Yang ZM. Genome-wide identification of Cd-responsive NRAMP transporter genes and analyzing expression of NRAMP 1 mediated by miR167 in brassica napus. Biometals. 2017;30:917–31.
49. Klousnitzer J, Xiang W, Polynice VM, Deslouches B. Comparative properties of helical and linear amphipathicity of peptides composed of arginine, tryptophan, and valine. Antibiotics. 2024;13(10):954.
50. Goki NH, Tehranizadeh ZA, Saberi MR, Khameneh B, Bazzaz BS. Structure, function, and physicochemical properties of pore-forming antimicrobial peptides. Curr Pharm Biotechnol. 2024;25(8):1041–57.
51. Ray S, Berry SP, Wilson EA, Zhang CH, Shekhar M, Singharoy A, Gaudet R. High-resolution structures with bound Mn2 + and Cd2 + map the metal import pathway in an Nramp transporter. Elife. 2023;12:e84006.
52. Verdonk H, Pivirotto A, Pavinato V, Hey J, Pond SLK. A new comparative framework for estimating selection on synonymous substitutions. bioRxiv. 2024:2024.2009.2017.613331.
53. Dhindsa RS, Wang Q, Vitsios D, Burren OS, Hu F, DiCarlo JE, et al. A minimal role for synonymous variation in human disease. Am J Hum Genet. 2022;109(12):2105–9.
54. Katneni U, Kimchi-Sarfaty C. Multiple mechanisms contribute to the phenotypic effects of synonymous variants. Hum Mutat. 2022;43(12):2324–5.
55. Liu S, Long T, Chen Z, Liu J, Cui W, Leng H, et al. Genome-wide identification of NRAMP family genes in *Populus trichocarpa* and their roles in transport of heavy metals. Tree Genet Genomes. 2023;19(6):51.
56. Yang Y, Zheng J, Liang Y, Wang X, Li K, Chen L, et al. Natural resistance-associated macrophage protein (Nramp) family in foxtail millet (*Setaria italica*): characterization, expression analysis and relationship with metal content under cd stress. Agronomy. 2023;13(8):2000.
57. Thomine S, Wang R, Ward JM, Crawford NM, Schroeder JI. Cadmium and iron transport by members of a plant metal transporter family in Arabidopsis with homology to Nramp genes. Proc Natl Acad Sci U S A. 2000;97(9):4991–6.
58. Nevo Y, Nelson N. The NRAMP family of metal-ion transporters. Biochimica et Biophysica Acta (BBA). 2006;1763(7):609–20.
59. Lanquar V, Lelièvre F, Bolte S, Hamès C, Alcon C, Neumann D, et al. Mobilization of vacuolar iron by AtNRAMP3 and AtNRAMP4 is essential for seed germination on low iron. EMBO J. 2005;24(23):4041–51.
60. Mäser P, Thomine S, Schroeder JI, Ward JM, Hirschi K, Sze H, et al. Phylogenetic relationships within cation transporter families of Arabidopsis. Plant Physiol. 2001;126(4):1646–67.
61. Dubos C, Stracke R, Grotewold E, Weisshaar B, Martin C, Lepiniec L. MYB transcription factors in Arabidopsis. Trends Plant Sci. 2010;15(10):573–81.
62. Wasternack C, Hause B. Jasmonates: biosynthesis, perception, signal transduction and action in plant stress response, growth and development. An update to the 2007 review in annals of botany. Ann Bot. 2013;111(6):1021–58.
63. Abe H, Urao T, Ito T, Seki M, Shinozaki K, Yamaguchi-Shinozaki K. *Arabidopsis* AtMYC2 (bHLH) and AtMYB2 (MYB) function as transcriptional activators in abscisic acid signaling. Plant Cell. 2003;15(1):63–78.
64. Wu D, Saleem M, He T, He G. The mechanism of metal homeostasis in plants: a new view on the synergistic regulation pathway of membrane proteins, lipids and metal ions. Membranes. 2021;11(12):984.
65. Hasan MK, Cheng Y, Kanwar MK, Chu X-Y, Ahammed GJ, Qi Z-Y. Responses of plant proteins to heavy metal stress—a review. Front Plant Sci. 2017;8:1492.
66. Biasini M, Bienert S, Waterhouse A, Arnold K, Studer G, Schmidt T, et al. SWISS-MODEL: modelling protein tertiary and quaternary structure using evolutionary information. Nucleic Acids Res. 2014;42(W1):W252-8.
67. Waterhouse A, Bertoni M, Bienert S, Studer G, Tauriello G, Gumienny R, Heer FT, de Beer TAP, Rempfer C, Bordoli L. SWISS-MODEL: homology modelling of protein structures and complexes. Nucleic Acids Res. 2018;46(W1):W296–303.
68. Lynch M, Conery JS. The evolutionary fate and consequences of duplicate genes. Science. 2000;290(5494):1151–5.
69. Tuskan GA, Difazio S, Jansson S, Bohlmann J, Grigoriev I, Hellsten U, et al. The genome of black cottonwood, Populus trichocarpa (Torr. & Gray). Science. 2006;313(5793):1596–604.

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